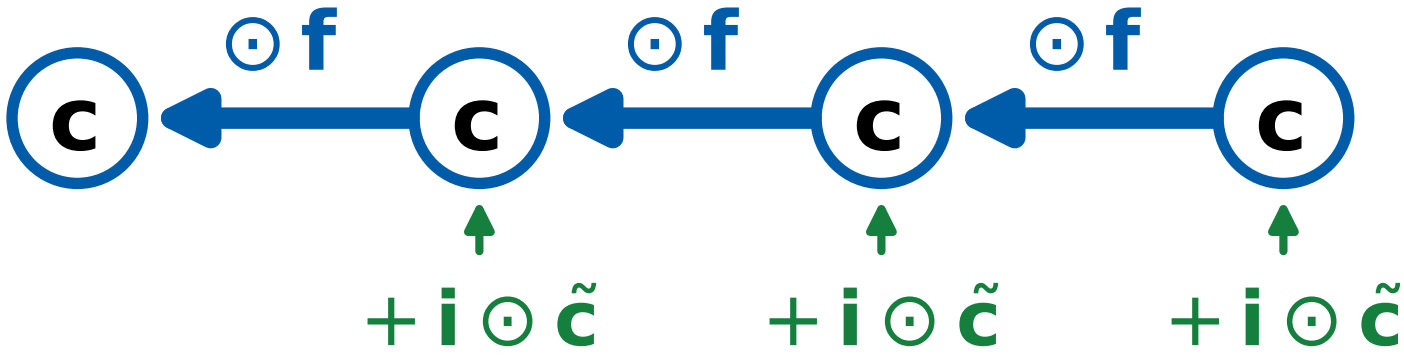
vanilla RNN

almost nothing arrives

LSTM cell

arrives intact if $\mathbf{f} \approx 1$